

Shared Substring Alignment

Identifying recurring structure within the encrypted GPS special messages

Group B — 10 shared characters, same day

2014-10-08	T	3	"	E	H	S	D	S	"	I	4	H	.	'	H	X	-	+	7	U	X		32 PRNs
2014-10-08	"	J	Y	3	2	Q	8	/	S	'	°	5	.	'	H	X	-	+	7	U	X		29 PRNs

Group D — 9 shared characters, 9 months apart

2021-06-05	Q	"	O	:	T	K	L	Y	4	7	I	R	P	1	6	A	C	0	2	4	B	A	31 PRNs
2020-09-19	O	H	6	°	S	3	L	Y	4	7	I	R	P	1	6	K		X	D	.	4	L	30 PRNs

Group A — 9 shared characters, 3 weeks apart

2019-12-11	2	C	K	0	E	6	:	U	'	5	P	S	F	8	T	J	9	0	F	D	X	'	31 PRNs
2019-11-20	Q	°	5	H	Y	1	:	U	'	5	P	S	F	8	T	7	8	J	9	7	I	Y	31 PRNs

Group C — 9 + 4 shared characters, 3 days apart

2024-04-07	4	3	P	A		:	Z	U	°	S	G	Z	8	P	J	D	L	-	:	I	M		32 PRNs
2024-04-10	:		2	N	K	M	Z	U	°	S	G	Z	8	P	J	E	G	B	:	I	M		32 PRNs

Group E — 7 shared characters, 3 months apart

2014-08-28	E	.	Y	S	/	Z	B	C	8	P	"	N	X	-	3	H	G	"	2	'	C	A	32 PRNs
2014-11-26	K	"	4	Y	C	5	T	L	5	-	U	5	9	T	H	H	G	"	2	'	C	A	32 PRNs

Group F — 7 + 1 shared characters, 3 months apart

2019-06-19	"	9	E	I	1	H	F	6	"	°	M	I	S	°	6	L	.	D	°	S	Z	B	31 PRNs
2019-09-14	V	L	V	5	G	K	'	0	V	9	M	7	S	°	6	L	.	D	°	F	'	H	32 PRNs

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

Byte position



Identical character at same position



Differing character